Instructions

Uppsala University

Department of Information Technology

Artificial Intelligence

2023-02-03

Instructions: Read through the complete exam and note below any unclear directives before you start solving the questions. **If you find any unclear directives, then please note the question number and explain what you think is unclear.**

This is a **closed book** exam. You may use a standard English dictionary and a calculator. You are NOT allowed to use any other aids (e.g. course material, notes, mobile phones). The exam is **individual**: you are NOT allowed to get help from other people to answer the questions. Cheating will be reported in accordance with university regulations.

Please notify an exam invigilator if you have questions about the exam. They will contact the course teacher (Andreína Francisco) if necessary.

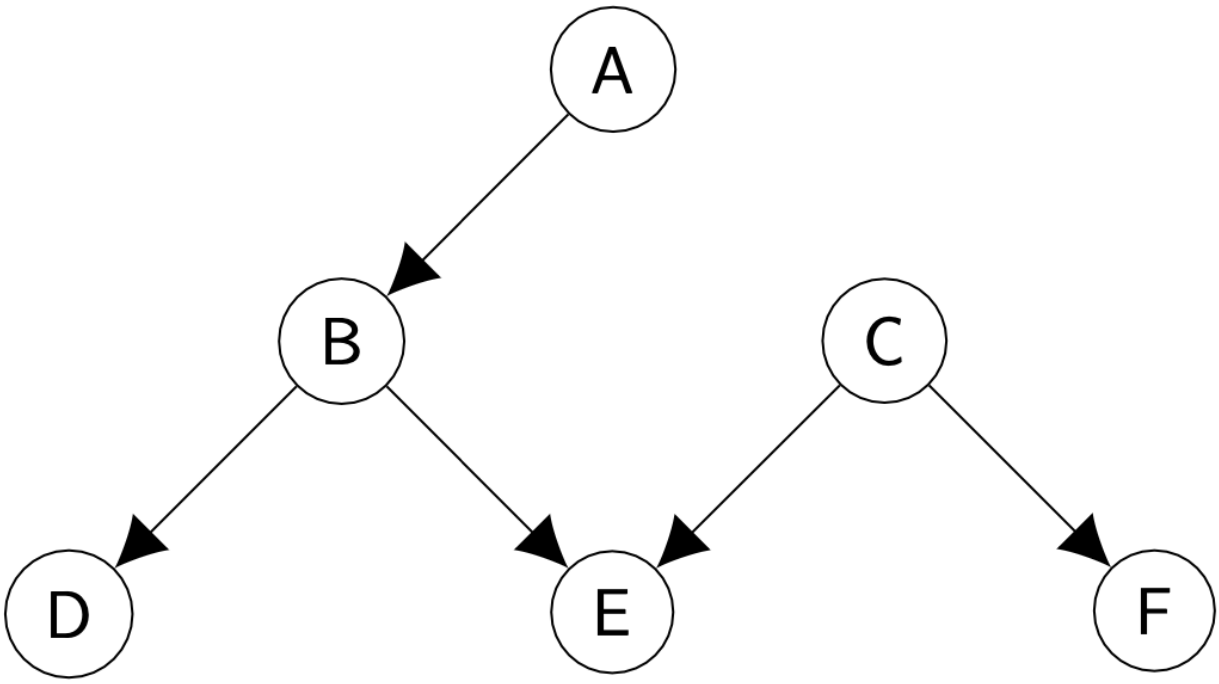
Grading scale. 0-49 pts. (U), 50-69 pts. (3), 70-84 pts. (4), 85-100 pts. (5)

There are 15 multiple-choice questions, each worth 2 points (30 points total). There are 7 problem-style questions, each worth 10 points (70 points total), some of which are divided into 2 parts.

Fill in your answer here

1 Bayesian Networks

Consider the following bayesian network:



Which probability distribution is associated with node B?
Select one alternative:

- P(B)
- P(B|A)
- P(A|B)
- P(B|A,D,E)
- P(B|D,E)

Maximum marks: 2

2 DFS

Which of the following is **not** a characteristic of DFS? (which statement is false)
Select one alternative:

- It traverses the depth of any particular path before exploring its breadth
- It cannot be used for cycle detection
- It can be implemented using recursion
- It has a complexity of O(V + E)
- It always gives the shortest path in a graph

Maximum marks: 2

3 A*

What happens when the heuristic function in the A* algorithm overestimates the cost?
Select one alternative:

- Nothing happens as long as the heuristic is monotonic
- The algorithm becomes faster
- The algorithm becomes slower
- The algorithm becomes uninformed
- The algorithm becomes non-optimal

Maximum marks: 2

4 Statistical Learning

Which of the following statements is true about statistical learning?
Select one alternative:

- When variables are correlated, it means there is a causal relationship
- None of the other statements are true
- Bayesian networks are a type of neural networks
- P(A,B) and P(A|B) both represent the joint distribution of A and B
- A distribution assigns probabilities to a random variable taking particular values

Maximum marks: 2

5 HMM

Which of the following is **not** a component of a Hidden Markov Model?
Select one alternative:

- Decision Tree
- All of the other alternatives are part of a HMM
- Emission probabilities
- Initial state probabilities
- State transition probabilities

Maximum marks: 2

6 Shortest Path Problems - 2023/10

Which of the following statements is true about Shortest Path Problems?
Select one alternative:

- Can only be solved using A*
- Can always be solved with BFS
- Can be used to find a route that minimises some cost
- Can only be solved if the graph is acyclic
- Are only applied to graphs that have costs associated with the nodes

Maximum marks: 2

7 alpha beta pruning (Q1V1)

Which of the following is required to be able to use alpha beta pruning?
Select one alternative:

- Zero-sum
- Turn-taking game
- Two players
- All of the other options
- Deterministic

Maximum marks: 2

8 Iterative Deepening

What is Iterative Deepening?
Select one alternative:

- A search strategy that combines the benefits of depth-first and breadth-first search
- An incomplete but fast search strategy
- A search strategy that combines depth-first search and heuristic search
- A search strategy that combines breadth-first search and uniform cost search
- A type of machine learning algorithm

Maximum marks: 2

9 PDDL

What is the main advantage of using PDDL?
Select one alternative:

- It simplifies the process of data analysis
- It simplifies the process of solving and ensures that a solution is found
- It simplifies the process of defining planning problems
- It simplifies the process of machine learning
- It simplifies the process of programming

Maximum marks: 2

10 Scheduling

Which of the following statements about scheduling is true?
Select one alternative:

- The goal is to provide a sequence of tasks needed to achieve a goal
- Resource-free scheduling is a way to estimate a lower-bound of the end time of the tasks
- It is not possible to obtain optimal solutions to scheduling problems
- The minimum-slack algorithm is never optimal
- The minimum-slack algorithm is always optimal

Maximum marks: 2

11 alpha beta pruning (Q2V1)

Which of the following is **not** a characteristic of alpha-beta pruning?
Select one alternative:

- It can only be used in two-player games with perfect information
- It is a greedy algorithm
- It is used in decision making
- It is a recursive algorithm
- It is used in game theory

Maximum marks: 2

12 Markov Blanket

What is a 'Markov Blanket' in a Bayesian Network?
Select one alternative:

- A mathematical formula used in the network to calculate a node's probability distribution
- The set of nodes around a given node
- The node’s parents, its children, and co-parents of its children
- A type of node
- A method of error correction

Maximum marks: 2

13 MiniMax (Q1V1)

What is the worst case time complexity of the Minimax algorithm?
Select one alternative:

- O(bd)
- O(n log n)
- O(b log d)
- O(n^2)
- O(b^d)

Maximum marks: 2

14 alpha beta pruning (Q3V1)

What does' pruning' mean in the context of the alpha-beta pruning algorithm?
Select one alternative:

- Finding the maximum value
- Finding the minimum value
- Increasing the search space
- Adding nodes to the tree
- Reducing the search space

Maximum marks: 2

15 Local Search

What is local search?
Select one alternative:

- A search strategy that only solves problems locally
- A search strategy that doesn't use heuristics
- A search strategy that incrementally improves the solution
- A search strategy only used in video games
- A search strategy that uses a global view of the problem

Maximum marks: 2

16 1DL340_Markov_Chains_2

Please read the PDF panel for the question text.

Fill in your answer here

Format

Words: 0

Maximum marks: 10

17 Scheduling - 1DL340

Consider the actions in the table below:

Index	Action	Duration	Uses	Consumes	After
0	Start	0			NA
1	Action 1	50		-1 cookie	0
2	Action 2	45	Plate	-1 cookie	0
3	Action 3	40	Plate		0
4	Action 4	5			1,2
5	Action 5	5	Cup	1 cookies	2,3
6	Action 6	25	Plate		4,5
7	Action 7	20	Plate, Cup	1 cookie	5
8	Finish	0			4,6,7

Provide a complete unconstrained schedule (5 points)

Fill in your answer here

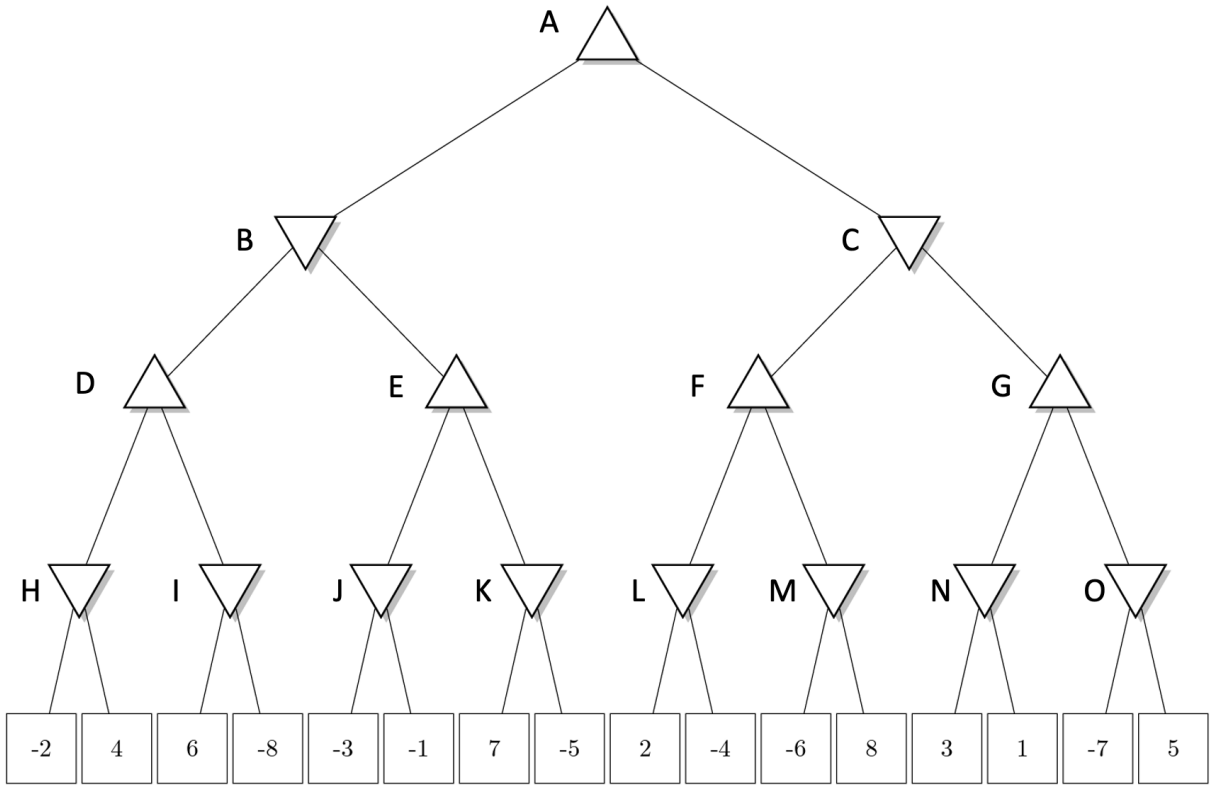
Provide a complete resource-constrained schedule (5 points)

Fill in your answer here

Maximum marks: 10

18 MiniMax - 1DL340 - 2024/02

Consider the following tree representing a zero-sum game, where triangles pointing up are max nodes, triangles pointing down are min nodes, and squares are terminal nodes with the corresponding value of utility function for max player.



Apply the mini max algorithm for finding the best action for the max player at the root. Uppercase letters represent the name of a given node.

Fill in your answer here

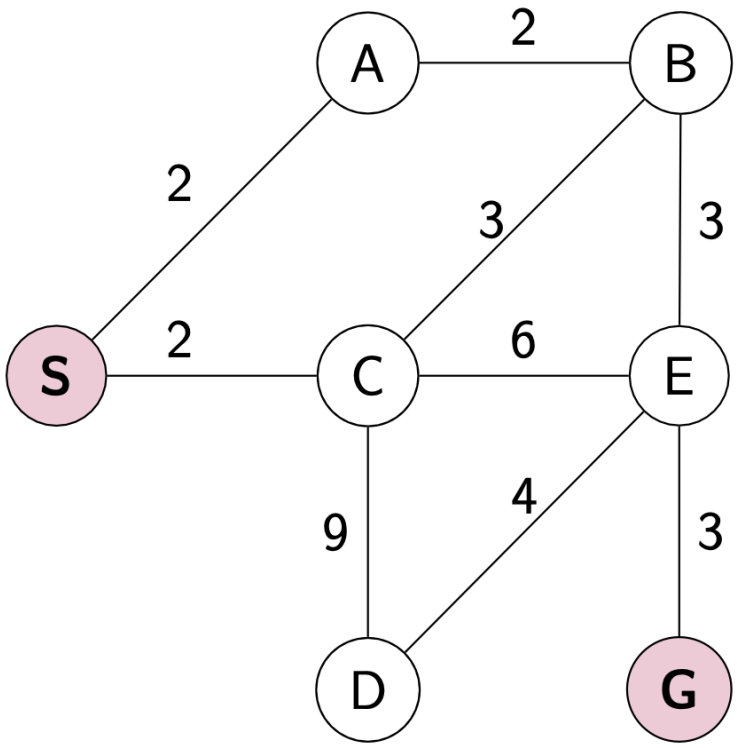
Format

Words: 0

Maximum marks: 10

19 Heuristic Search - 1DL340 - 2024/02

Use A* to find a shortest path from the start node S to the goal node G in the graph below:



Provide an admissible heuristic (2 points)

Fill in your answer here

Show all your workings (all the steps in the algorithm to find the goal) (8 points)

Fill in your answer here

Maximum marks: 10

20 Planning - 1DL340

What is PDDL? Explain all components of a PDDL problem. Be as precise as possible and give a small example. You do not have to worry about the exact syntax of PDDL expressions, but you should have all the components of PDDL expressions. In your small example explain what each syntactic element does. (10 points)

Fill in your answer here

Format

Words: 0

Maximum marks: 10

21 Guest lectures - 1DL340

How are the goals of AI in computer games different from the typical goals of more 'academic' AI? (5 points)

Fill in your answer here

Ant Colony Optimization is a natural computation method inspired by the behaviour of ants and termites. The core concept which is modelled here is called stigmergy. What is stigmergy? Explain the concept and why this way to solve problems is interesting to computer scientists. (5 points)

Fill in your answer here

Maximum marks: 10

22 Philosophy, ethics, and safety of AI

You have a job as an ethical adviser for a major company that runs a popular social media site. You are working with the team that is developing the AI to manage recommendations for users' newsfeeds.

Discuss the following two core functions related to the design of the newsfeed AI:

- Identifying ethical issues of AI, and
- Ensuring the accountability of AI.

For both of these core functions, you should consider the Teleological perspective (that is, considers goods or harms to individuals or society).

Your answer should be of a minimum of about 150 words on each core function.

Fill in your answer here

Format

Words: 0

Maximum marks: 10

Question 16
Attached



The question has two parts. You are to answer both parts, and make it clear in your answer where each part starts.

Part A

Consider a Markov chain with two states S (Snow) I (Ice) with the following transition matrix.

$$\begin{pmatrix} P(S_{i+1} \mid S_i) = 0.9 & P(I_{i+1} \mid S_i) = 0.1 \\ P(S_{i+1} \mid I_i) = 0.5 & P(I_{i+1} \mid I_{i+1}) = 0.5 \end{pmatrix},$$

The initial probability are $P(S_0) = 0.5$ and $P(I_0) = 0.5$.

Derive an expression for the probability that it is snowing after 4 days $P(S_4)$. You need not give the actual numerical answer, but the sequence of multiplications that give that answer.

Part B

You are running a coffee shop. Because you work so hard you cannot see the weather outside. You just observe the number of customers that you have. You are too busy to count how many customers you have, so you have three estimates F (Few) , H (Half full), P , (Packed with many people). You are trying to estimate the probability that it is raining or sunny.

Your Hidden Markov Model has three observations (or emissions): F, H, and P, and two hidden states R (Rainy), S (Sunny). Given the initial probabilities of it being raining $P(R_0)$ or sunny $P(S_0)$. Derive an algebraic expression that is efficient to compute for the probability that it is raining given that you have observed the sequence F, H, P.